Scenario-1: John is planning to invest in a retirement plan that offers a guaranteed return. He is considering two options for his investment:

1. **Option 1:** Invest $10,000 today in a retirement plan that guarantees an annual return of 5% for the next 10 years.
2. **Option 2:** Invest $12,000 in the same plan after 5 years (starting from the sixth year) but receive the same 5% annual return for the following 5 years (years 6–10).

John wants to know which option will provide him with a higher future value after 10 years.

Solution Explanation:

To compare the two options, we need to calculate the **future value (FV)** of both investments using the **time value of money (TVM)** concept.

The Future Value formula is:

$$FV = PV(1 + i)^n$$

Where:

- **FV = Future Value**
- **PV = Present Value (initial investment)**
- **i = Annual interest rate (5% or 0.05)**
- **n = Number of periods (years)**

Option 1: Invest $10,000 today for 10 years

For Option 1, the present value (PV) is $10,000, the interest rate (r) is 5%, and the number of periods (n) is 10 years.

$$FV_1 = PV(1 + k)^n$$
$$= 10{,}000(1 + .05)^{10}$$
$$= 10{,}000 \times 1.62889$$
$$= 16{,}288.9$$

The future value of the investment in Option 1 will be **$16,288.9** after 10 years.

Option 2: Invest $12,000 after 5 years for 5 years

For Option 2, John invests $12,000 after 5 years, meaning that for the first 5 years, the money earns no return. Once the investment is made, the same 5% return applies for the next 5 years.

$$FV_2 = PV(1 + k)^n$$
$$= 12{,}000(1 + .05)^5$$
$$= 12{,}000 \times 1.27628$$
$$= 15{,}315.36$$

Conclusion:

- **Option 1 (Invest $10,000 today)** provides a **future value of $16,288.9**.
- **Option 2 (Invest $12,000 after 5 years)** provides a **future value of $15,315.36**.

John would receive a higher return by choosing **Option 1**. The key insight is that investing earlier (even with a smaller amount) allows compounding to work for a longer period, resulting in a higher future value.

This case study demonstrates the **importance of the time value of money**, showing how money invested earlier grows more due to compounding over time.

Scenario-2: Lisa wants to save for her child's college education, which will begin 15 years from now. She plans to make annual deposits of **$5,000** at the end of each year into an investment account that earns a fixed interest rate of **6%** per year.

Lisa is curious to know how much money will be available in the account when her child starts college after 15 years. She wants to know the future value of the annuity (the series of annual payments) at that time.

Solution Explanation:

To solve this problem, we will use the **Future Value of an Annuity (FVA)** formula, which calculates the future value of a series of regular payments (annuity).

The formula for the **future value of an ordinary annuity** (payments made at the end of each period) is:

$$FVA_t = PMT\left[\frac{(1+r)^t - 1}{r}\right]$$

Where:

- **FVA = Future Value of the annuity**
- **Pmt = Payment amount ($5,000)**
- **r = Interest rate per period (6% or 0.06)**
- **n = Number of periods (15 years)**

$$\textbf{FVA} = 5{,}000\left[\frac{(1+0.06)^{15}-1}{0.06}\right]$$

$$=5000*23.276$$

Conclusion:

After 15 years of making annual payments of $5,000 at a 6% interest rate, Lisa will have **$116,380** in the investment account.

This case study demonstrates the power of **compounding in an annuity**, where regular contributions grow significantly over time due to the interest compounding on both the principal and accumulated interest.
The longer the time period and the higher the interest rate, the larger the future value of the annuity becomes.

Scenario-3: Michael is planning to save for a down payment on a house he plans to buy in 10 years. He has determined that he needs **$100,000** by the end of 10 years. Michael will make equal annual deposits into an investment account that earns **8% interest** per year.
He wants to know how much he needs to deposit **at the end of each year** in order to accumulate $100,000 at the end of 10 years.

Solution Explanation:

This is a **Future Value of an Annuity (FVA)** problem where we need to calculate the annual payment amount PPP needed to reach a target future value of $100,000.
The **Future Value of an Annuity** formula is:

$$FVA_t = PMT\left[\frac{(1+r)^t - 1}{r}\right]$$

Where:

- **FVA** = Future Value of the annuity ($100,000)
- **PMT** = Payment amount (to be calculated)
- **r** = Interest rate per period (8% or 0.08)
- **n** = Number of periods (10 years)

We need to solve for **PMT** , the annual payment.

$$\textbf{\$100,000} = \textbf{PMT}\left[\frac{(1+0.08)^{10}-1}{0.08}\right]$$

$$\textbf{PMT= 6,903.89}$$

Conclusion:

Michael will need to deposit $6,903.89 at the end of each year for the next 10 years in order to accumulate $100,000 at the end of the 10-year period, assuming an 8% annual return.

Explanation:

This case study highlights the use of the **Future Value of an Annuity** formula to find the required regular payment. The time value of money shows that, due to compounding interest, Michael can save a smaller amount annually compared to

saving the entire $100,000 in one lump sum. The longer the period and the higher the interest rate, the smaller the required annual payment to achieve the future goal.

MATH-1:

What is the Present Value of the following Cash Flows at an Interest rate of 12% per year?

I. $100 received today?
II. $100 received after 5 Years from now?
III. $100 received each year beginning 1 year from now and ending 10 years from now?

MATH-2:

What is the value of the following investments: 5 years from now at an Interest rate of 8% per year depending on the Compounding Period?

I. $100 invested today with interest Compounded Annually.
II. $100 invested today with interest Compounded Quarterly.
III. $100 invested today with interest Compounded Monthly.

MATH-3:

Find the Future Value of the following investment at an Interest rate of 8% per year, Compounded Annually.

I. $100 is invested each year beginning 1 year from now and continues through Year 10, when the Proceeds are withdrawn.
II. $100 is invested each year starting today and continuing through Year 10, when the Proceeds are withdrawn.
III. $100 is invested each year beginning 1 year from now and continues through Year 9, when the Proceeds are to be withdrawn in year 10.

MATH-4:

If you put $100 into a Savings Certificate today paying an Interest rate of 10% per year. How much money will you have in the account after 6 years if no withdrawals are made from it until then, how much of this is Interest?

MATH-5:

If you wish to have $1000 in 10 years from now, how much money must you invest today in a Savings Certificate that pays 8% interest per year?

MATH-6:

If you wish to have $1000 in a Savings Account after 10 years from now an Interest rate of 10%, how much money will you invest in your account each year starting 1 year from now?

MATH-8:

The DELCO Company is attempting to determine the most it should pay to purchase a particular Annuity. The firm requires a minimum return of 8% on all investments. The Annuity consists of cash flows of $700 per year for 5 years.